

Sharp Berry–Esseen bounds for Bernoulli sums and reductions for general arrays

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Abstract

Independent centered two-point summands with a common support diameter satisfy the Berry–Esseen inequality with $C_E = (3 + \sqrt{10})/(6\sqrt{2\pi})$. We prove that this constant is optimal in the common-diameter class. We also prove the strict inequality for two centered two-point summands with arbitrary diameters. The common-diameter proof uses Schulz’s binomial theorem, optimization over success probabilities, and validated finite calculations. Separate results give atom and short-interval bounds, with an equality condition on the diameters. For general arrays, we prove a variance-preserving reduction to summands with at most three support points and an attained quadratic dual formula. We then reduce the unrestricted sharp bound to two open problems: the bound for Bernoulli arrays with arbitrary diameters and an inequality for three-point mixtures.

1 Results

Let $X_1, \dots, X_n$ be independent centered real random variables with finite absolute third moments. Throughout, write

$$B^2 = \sum_i \sigma_i^2 > 0, \quad \sigma_i^2 = \mathbb{E}X_i^2, \quad \ell = B^{-3} \sum_i \mathbb{E}|X_i|^3, \quad (1.1)$$

$$\Delta = \sup_{x \in \mathbb{R}} \left| \mathbb{P}\{B^{-1} \sum_i X_i \leq x\} - \Phi(x) \right|.$$

Here Φ and ϕ are the standard normal distribution function and density. The independent Berry–Esseen problem asks for the least universal constant in $\Delta \leq C\ell$. Esseen’s lower bound is

$$C_E = \frac{3 + \sqrt{10}}{6\sqrt{2\pi}}, \quad K_E = \sqrt{2\pi}C_E = \frac{3 + \sqrt{10}}{6}. \quad (1.2)$$

The equality of the unrestricted constant with C_E remains open.

Define

$$t_* = \sqrt{10} - 3, \quad p_E = \frac{1 - t_*}{2} = \frac{4 - \sqrt{10}}{2}, \quad q_E = 1 - p_E.$$

The variance-one Esseen law, in the orientation used here, is

$$\mu_E = p_E \delta_{\sqrt{q_E/p_E}} + q_E \delta_{-\sqrt{p_E/q_E}}. \quad (1.3)$$

Its third moment is positive. Esseen’s i.i.d. sequence with this law has $\Delta/\ell \rightarrow C_E$ [5].

Theorem 1.1 (A common support diameter). *Suppose each X_i has at most two support points and there is an $h > 0$ such that every nondegenerate X_i has support diameter exactly h . Then $\Delta \leq C_E \ell$. The constant C_E is optimal over this class.*

Theorem 1.2 (Two arbitrary support diameters). *For two independent centered random variables, each supported on at most two points and with positive total variance, $\Delta < C_E \ell$.*

Theorem 1.2 does not identify the optimal constant for two summands. For three or more summands, the class with unequal diameters remains unresolved, even when all diameters are integer multiples of a common positive number.

The binomial input is Schulz's theorem [9, Theorem 1]: for $N \sim \text{Bin}(n, p)$, $n \geq 1$, $0 < p < 1$, and $q = 1 - p$,

$$\sup_x |\mathbb{P}\{(N - np)/\sqrt{npq} \leq x\} - \Phi(x)| < C_E \frac{p^2 + q^2}{\sqrt{npq}}. \quad (1.4)$$

Our common-diameter theorem extends this assertion to unequal success probabilities. Its proof reduces probability vectors first to two distinct values, then to a binomial sum and one exceptional Bernoulli variable. Section 7 describes the computational supplement. Sections 5 and 6 give the reductions for general arrays. The atom and interval estimates in Section 4 are separate results.

He and Cheng [6] prove the sharp bound for sufficiently large i.i.d. samples. The companion manuscript [1] adapts their approach to general independent arrays with sufficiently small Lyapunov ratio. Its proof uses Theorem 1.1 to bound the common-diameter Bernoulli array obtained in its local comparison. The finite results proved here do not require that asymptotic argument.

For a centered sum S of variance B^2 , write

$$F_S^+(y) = \mathbb{P}(S \leq y), \quad F_S^-(y) = \mathbb{P}(S < y), \quad H_S^\varepsilon(y) = C_E \ell + \varepsilon \Phi(y/B) - \varepsilon F_S^\varepsilon(y), \quad \varepsilon \in \{+1, -1\}.$$

The notation F_S^ε selects the closed CDF for $\varepsilon = +1$ and its left limit for $\varepsilon = -1$. Continuity of Φ gives

$$\Delta = \sup_y \max\{F_S^+(y) - \Phi(y/B), \Phi(y/B) - F_S^-(y)\}. \quad (1.5)$$

For finite support it suffices to inspect these two values at each atom: between atoms, $F_S^+ - \Phi$ decreases and $\Phi - F_S^+$ increases. Reflection gives $H_S^-(y) = H_{-S}^+(-y)$.

2 Proof for a common support diameter

Let $B_1, \dots, B_n$ be independent Bernoulli variables with parameters $p_1, \dots, p_n$. Put $q_i = 1 - p_i$ and

$$N_p := \sum_{i=1}^n B_i, \quad \mu := \sum_{i=1}^n p_i, \quad V(p) := \sum_{i=1}^n p_i q_i, \quad \mathcal{B}(p) := \sum_{i=1}^n p_i q_i (p_i^2 + q_i^2). \quad (2.1)$$

Write $F_j(p) = \mathbb{P}(N_p \leq j)$, with $F_j = 0$ below the support and $F_j = 1$ above it. For an integer k and $V(p) > 0$, define the closed and strict deficits

$$\mathcal{H}_+(p; k) := C_E \frac{\mathcal{B}(p)}{V(p)^{3/2}} + \Phi\left(\frac{k - \mu}{\sqrt{V(p)}}\right) - F_k(p), \quad (2.2)$$

$$\mathcal{H}_-(p; k) := C_E \frac{\mathcal{B}(p)}{V(p)^{3/2}} - \Phi\left(\frac{k - \mu}{\sqrt{V(p)}}\right) + F_{k-1}(p). \quad (2.3)$$

By (1.5), the common-diameter bound is equivalent to the nonnegativity of both deficits for every integer k . Complementation gives the exact reflection

$$\mathcal{H}_-(p_1, \dots, p_n; k) = \mathcal{H}_+(q_1, \dots, q_n; n - k). \quad (2.4)$$

Thus it is enough, when convenient, to treat the closed CDF value.

We use the following bound for third differences of a Poisson-binomial mass function.

Lemma 2.1 (A bound for third differences). *Let $c_j = \mathbb{P}(C = j)$ be the mass function of any finite Poisson-binomial law, extended by zero off its support, and let*

$$W = \text{Var } C, \quad \Delta^3 c_j := c_j - 3c_{j-1} + 3c_{j-2} - c_{j-3}.$$

Then

$$\sup_j |\Delta^3 c_j| < \frac{8C_E}{(W+1)^{3/2}}. \quad (2.5)$$

Proof. Set $A = 8C_E$. The rational estimates $\sqrt{10} > 79/25$ and $\pi < 22/7$ give $A > 16/5$.

Suppose first that $0 \leq W \leq 1/4$. Write $C = \sum_i B_i$, where the B_i are independent with $B_i \sim \text{Bern}(p_i)$, and put $q_i = 1 - p_i$. Choose the more likely value of each B_i as a base configuration, put $\varepsilon_i = \min(p_i, q_i)$, and let J count deviations from that configuration. If $E = \sum_i \varepsilon_i$ and $P_r = \mathbb{P}(J = r)$, then

$$E \leq 2W \leq \frac{1}{2}, \quad P_0 \geq 1 - E \geq \frac{1}{2}.$$

With $R = \sum_i \varepsilon_i / (1 - \varepsilon_i)$, one has $W \leq E \leq R \leq 2E \leq 1$, $P_1 = P_0 R$, and

$$P_0 \geq e^{-R}, \quad P_1 \geq R e^{-R} \geq W e^{-W}.$$

Let m be the number of ones in the base configuration. At the two central indices $m+1$ and $m+2$, the no-deviation contribution to $\Delta^3 c_j$ has magnitude three. Every one-deviation contribution has the opposite sign and magnitude at least one; every remaining contribution has magnitude at most three. Since $P_0 \geq 1/2$, the central sign cannot reverse, and hence

$$|\Delta^3 c_j| \leq 3P_0 - P_1 + 3\mathbb{P}(J \geq 2) = 3 - 4P_1 \leq 3 - 4W e^{-W} \quad (2.6)$$

at those two indices. At every other index, the base contribution has magnitude at most one, so

$$|\Delta^3 c_j| \leq P_0 + 3(1 - P_0) \leq 1 + 4W. \quad (2.7)$$

The outer bound in (2.7) is at most two, whereas $A(1+W)^{-3/2} > 2$ on this interval. For the central bound (2.6), use $e^{-W} \geq 1 - W$ and

$$(1+W)^{-3/2} \geq h(W) := 1 - \frac{3}{2}W + \frac{15}{8}W^2 - \frac{35}{16}W^3.$$

For

$$P(W) := Ah(W) - (3 - 4W + 4W^2),$$

one has $h'(W) < 0$ and, because $A > 3$,

$$P'(W) < 3h'(W) + 4 - 8W = -\frac{1}{2} + \frac{13}{4}W - \frac{315}{16}W^2 < 0.$$

At the right endpoint,

$$P(1/4) = A \frac{725}{1024} - \frac{9}{4} > \frac{1}{64}.$$

This proves (2.5) for $W \leq 1/4$.

For $W \geq 1/4$, Fourier inversion and

$$|1 - p + pe^{it}| \leq \exp\{-2p(1-p)\sin^2(t/2)\}$$

give

$$|\Delta^3 c_j| \leq I(W) := \frac{16}{\pi} \int_0^{\pi/2} \sin^3 x e^{-2W \sin^2 x} dx. \quad (2.8)$$

Put $J_r(W) = \int_0^{\pi/2} \sin^r x e^{-2W \sin^2 x} dx$. Integration of the derivative of $\sin^4 x \cos x e^{-2W \sin^2 x}$ yields

$$(5 + 4W)J_5 = 4J_3 + 4WJ_7, \quad \frac{J_5}{J_3} \geq \frac{4}{5 + 4W}.$$

Consequently

$$\frac{d}{dW} \log\{I(W)(1+W)^{3/2}\} = -2\frac{J_5}{J_3} + \frac{3}{2(1+W)} \leq -\frac{8}{5+4W} + \frac{3}{2(1+W)} < 0.$$

It remains to check $W = 1/4$. The bound $e^{-u} \leq 1 - u + u^2/2$ gives

$$I(1/4) \leq \frac{16}{\pi} \left(J_3(0) - \frac{1}{2}J_5(0) + \frac{1}{8}J_7(0) \right) = \frac{256}{35\pi}.$$

Finally,

$$\frac{256}{35\pi} < \frac{A}{(5/4)^{3/2}},$$

because, after squaring positive quantities, this is $5760 < 49\pi(19 + 6\sqrt{10})$, which follows from $\pi > 157/50$ and $\sqrt{10} > 79/25$. The normalized majorant in (2.8) is decreasing, completing the proof. $\square$

Proposition 2.2 (Reduction to a binomial sum and one Bernoulli variable). *Fix n , a mean μ , an integer k , and either deficit in (2.2)–(2.3). If that deficit is negative somewhere on*

$$\mathcal{P}_{n,\mu} := \left\{ p \in [0,1]^n : \sum_{i=1}^n p_i = \mu \right\},$$

then, after deleting deterministic coordinates and possibly applying the reflection (2.4), a negative global minimum has the form

$$N_p \sim \text{Bin}(M, p) + \text{Bern}(x), \quad M \geq 1, \quad 0 < p, x < 1, \quad p \neq x. \quad (2.9)$$

Proof. Since $\mathcal{B}(p) \geq V(p)/2$, extend the selected deficit by $+\infty$ on $V = 0$. It then has a global minimum away from the zero-variance boundary. Choose a face of least dimension on which the selected deficit is negative somewhere. Among its global minimizers, choose one maximizing $\sum_i p_i^3$. A coordinate at zero can be deleted. A coordinate at one can be deleted after replacing (μ, k) by $(\mu - 1, k - 1)$. Both operations preserve $k - \mu$ and the selected deficit. Each face is identified bijectively with the reduced polytope, so global minimality is preserved; the cubic tie-break is

preserved as well, because the deleted cube is the constant zero or one on that face. Minimal dimension therefore makes the chosen minimizer interior.

Select three coordinates and write e_1, e_2, e_3 for their elementary symmetric polynomials. Their generating polynomial is

$$\prod_{i=1}^3 \{1 + p_i(w-1)\} = 1 + e_1(w-1) + e_2(w-1)^2 + e_3(w-1)^3. \quad (2.10)$$

Holding e_1, e_2 fixed preserves the mean and the triple variance $V_3 = e_1 - e_1^2 + 2e_2$. At every fixed integer threshold, the CDF of the full sum is affine in e_3 , and the contribution of the selected triple to $\mathcal{B}$ is

$$\begin{aligned} \mathcal{B}_3 = & e_1 - 3e_1^2 + 4e_1^3 - 2e_1^4 + (6 - 12e_1 + 8e_1^2)e_2 - 4e_2^2 \\ & + 4(3 - 2e_1)e_3. \end{aligned} \quad (2.11)$$

Thus the whole deficit is affine on the feasible e_3 interval. If the three roots are distinct and interior, the current e_3 is interior. A nonzero slope permits a decrease of the deficit; a zero slope permits an increase of e_3 and therefore of

$$p_1^3 + p_2^3 + p_3^3 = e_1^3 - 3e_1e_2 + 3e_3.$$

Both alternatives contradict the choice of minimizer. Hence it has at most two distinct interior probabilities.

Schulz's theorem (1.4) excludes a negative deficit when all probabilities are equal. Apply the reflection (2.4) if necessary, so the selected deficit is closed. Suppose the two values are $0 < a < b < 1$, with multiplicities r and s . It remains to show that $r, s \geq 2$ is impossible. For a selected triple, let d_j be the mass function of the other coordinates and put $\Delta^2 d_j = d_j - 2d_{j-1} + d_{j-2}$. By (2.10), the coefficient of e_3 in the selected CDF is $-\Delta^2 d_k$. Hence the coefficient of e_3 in the deficit is

$$L(e_1; d) = 4\gamma(3 - 2e_1) + \Delta^2 d_k, \quad \gamma := \frac{C_E}{V^{3/2}}. \quad (2.12)$$

For $f(t) = t^3 - e_1 t^2 + e_2 t$, one has $f''(a) = 2(a - b) < 0$ at (a, a, b) and $f''(b) = 2(b - a) > 0$ at (a, b, b) . Thus the values of e_3 at these triples are, respectively, the upper and lower endpoints of the interval for which $t^3 - e_1 t^2 + e_2 t - e_3$ has three roots in $[0, 1]$. Global minimality gives

$$L_{aab} \leq 0 \leq L_{abb}. \quad (2.13)$$

If the other endpoint of this interval first sends a root to zero or one, the same one-sided endpoint conditions hold; no interior endpoint has been omitted.

When $r, s \geq 2$, delete two copies of each parameter and let c_j be the mass function of the remaining core. The deletions associated with the two triples are $c * \text{Bern}(b)$ and $c * \text{Bern}(a)$. Direct substitution in (2.12) gives

$$L_{abb} - L_{aab} = (b - a)\{-8\gamma + \Delta^3 c_k\}. \quad (2.14)$$

The slope-difference identity (2.14) and the signs in (2.13) would force

$$\Delta^3 c_k \geq \frac{8C_E}{V^{3/2}}.$$

If W is the variance of the core, then $V = W + 2a(1 - a) + 2b(1 - b) \leq W + 1$. Lemma 2.1 gives the opposite inequality. Thus one of r, s equals one, which is exactly the family (2.9). $\square$

For a binomial sum and one Bernoulli variable, three adjacent binomial masses bound the second derivative of the deficit. The following inequality will exclude a negative minimum when the binomial sum has at least two summands. The two-summand case is proved in Section 3.

Lemma 2.3 (A curvature inequality on a bounded domain). *Let $0 < p, x < 1$, $q = 1 - p$, and*

$$pq \leq W \leq 3, \quad 0 \leq f \leq 1, \quad A_0 := W - (1 - f)q^2 \geq 0.$$

Set

$$\begin{aligned} B_0 &:= p^2 + (1 - f)(W + 2pq), & V &:= x(1 - x) + W + pq, \\ t_x &:= 1 - 2x, & t_p &:= 1 - 2p, & y &:= \frac{f(W + pq)}{1 - qf}, & z &:= \frac{y - x}{\sqrt{V}}. \end{aligned}$$

Then

$$\frac{2C_E}{\sqrt{V}} \{4W + 2 + t_x t_p\} + z\phi(z)(2 + z^2) - 5V \frac{fB_0}{(2 - f)B_0 + A_0} > 0. \quad (2.15)$$

Proof. For $V \leq 1/5$, put $S_2 = t_x^2 + t_x t_p + t_p^2$. The inequalities $V \geq \{2 - t_x^2 - t_p^2\}/4$ and $S_2 \geq (t_x^2 + t_p^2)/2$ imply $S_2 \geq 1/3$. Since $4V + S_2 = 4W + 2 + t_x t_p$, the arithmetic-geometric mean inequality bounds the first term in (2.15) from below by

$$C_E \left(8\sqrt{V} + \frac{2S_2}{\sqrt{V}} \right) \geq \frac{8C_E}{\sqrt{3}}.$$

The normal contribution is at least $-4/(e\sqrt{\pi})$, since $u(2 + u^2)\phi(u)$ is maximized at $u = \sqrt{2}$. Moreover

$$(2 - f)B_0 + A_0 - fB_0 = 2(1 - f)B_0 + A_0 \geq 0,$$

so the neighbor fraction is at most one and its entire subtracted contribution is at most $5V \leq 1$. The rational bounds

$$C_E > \frac{2}{5}, \quad \sqrt{3} < \frac{26}{15}, \quad e > \frac{19}{7}, \quad \sqrt{\pi} > \frac{7}{4}$$

give the strict margin

$$\frac{8C_E}{\sqrt{3}} - \frac{4}{e\sqrt{\pi}} - 1 > \frac{24}{13} - \frac{16}{19} - 1 = \frac{1}{247}. \quad (2.16)$$

For the complement $V \geq 1/5$, the normal term is bounded below by

$$z\phi(z)(2 + z^2) \geq \sqrt{\frac{2}{\pi}} \min\{z, 0\}. \quad (2.17)$$

Indeed the claim is immediate for $z \geq 0$; for $z = -u < 0$, the function $(2 + u^2)\phi(u)$ decreases because its derivative is $-u^3\phi(u)$, and its value at zero is $2\phi(0) = \sqrt{2/\pi}$. We verify the remaining domain using 192-bit Arb intervals and the following three parametrizations:

- (i) $p = (1 + s_p)/2$, $W = pq + s_W(3 - pq)$, $f = s_f$;
- (ii) $p = s_p/2$, $W = q^2 + s_W(3 - q^2)$, $f = s_f$;
- (iii) $p = s_p/2$, $W = pq + s_W(q^2 - pq)$, $f = 1 - (1 - s_f)W/q^2$,

where $s_p, s_W, s_f, x \in [0, 1]$. The first chart is exactly $p \geq 1/2$. For $p \leq 1/2$, the other two charts split at $W = q^2$; below that interface, $A_0 \geq 0$ is equivalent to $f \geq 1 - W/q^2$, which is precisely the range of the third parametrization. The interfaces overlap. Thus the charts cover the whole stated domain. In the third chart $A_0 = s_f W$; the first two charts give, respectively, $A_0 = q s_p + s_W(3 - pq) + f q^2$ and $A_0 = s_W(3 - q^2) + f q^2$. On each box the program uses a downward enclosure for the smooth term, the lower bound (2.17) for the normal term, and an upward enclosure for the subtracted mass term in (2.15). A box is accepted only when the resulting outward lower endpoint is positive. Otherwise the widest normalized coordinate is bisected dyadically; an unresolved box at the prescribed depth or box budget aborts the program. Boxes touching both $p = 0$ and $f = 1$ use the weaker limit $y \geq 0$ rather than dividing by an interval containing zero. If interval dependency does not certify the three-neighbor denominator as positive, the program does not divide; it uses the universal bound of one for the neighbor fraction and accepts only if the resulting lower gap is positive. The finite cover and its strict acceptance bounds are recorded in Section 7.

The verifier is `verify/finite/common_diameter/singleton_certificate.py`. The supporting records are listed in `certificates/finite/`. $\square$

Proposition 2.4 (A binomial sum and one Bernoulli variable). *For every $M \geq 1$, $0 < p, x < 1$, $p \neq x$, and every integer k , both deficits of $\text{Bin}(M, p) + \text{Bern}(x)$ are nonnegative.*

Proof. By (2.4), it suffices to treat the closed CDF value. Suppose it has a negative deficit. Proposition 2.2, applied with the same mean and threshold, gives a negative global minimum of the same form with $p \neq x$. Relabel its parameters as M, p, x .

First suppose $M \geq 2$, put $m = M - 1$, and let

$$D \sim \text{Bin}(m, p), \quad d_j = \mathbb{P}(D = j), \quad W = mpq.$$

Vary the success probabilities of the exceptional variable and one variable in the repeated group, keeping their sum fixed. Write their probabilities as $\bar{p} + d$ and $\bar{p} - d$, and put $u = d^2$. Their generating polynomial is

$$\{1 + \bar{p}(w - 1)\}^2 - u(w - 1)^2,$$

so the selected closed CDF along the smoothing curve satisfies the exact affinity

$$\mathcal{F}_k(u) = \mathcal{F}_k(0) + (d_{k-1} - d_k)u. \quad (2.18)$$

At the current point set

$$\begin{aligned} t_x &= 1 - 2x, & t_p &= 1 - 2p, & \xi &= d_{k-1} - d_k, \\ S_2 &= t_x^2 + t_x t_p + t_p^2, & z &= \frac{k - x - Mp}{\sqrt{V}}, & V &= x(1 - x) + Mpq. \end{aligned}$$

The moving pair satisfies

$$V' = -2, \quad \mathcal{B}' = -2S_2, \quad \mathcal{B}'' = -8.$$

If

$$\mathcal{A}(u) = C_E \frac{\mathcal{B}(u)}{V(u)^{3/2}} + \Phi\left(\frac{k - \mu}{\sqrt{V(u)}}\right),$$

then $\mathcal{H}_+ = \mathcal{A} - \mathcal{F}_k$. Since the current unequal pair is an interior point of the u interval, stationarity and (2.18) give $\mathcal{A}' = \xi$, while $\mathcal{H}_+'' = \mathcal{A}''$. Direct differentiation gives

$$\begin{aligned}\mathcal{A}' &= C_E \frac{\mathcal{B}'V + 3\mathcal{B}}{V^{5/2}} + \frac{z\phi(z)}{V}, \\ \mathcal{A}'' &= C_E \frac{\mathcal{B}''V^2 + 6\mathcal{B}'V + 15\mathcal{B}}{V^{7/2}} + \frac{z\phi(z)(3 - z^2)}{V^2}.\end{aligned}$$

Eliminating $\mathcal{B}$ from these two identities with $\mathcal{A}' = \xi$ gives the exact curvature formula

$$V^2 \mathcal{H}_+'' = 5V\xi - C_E \left(8\sqrt{V} + \frac{2S_2}{\sqrt{V}} \right) - z\phi(z)(2 + z^2). \quad (2.19)$$

A local minimum would require the left side to be nonnegative.

Extend d_j by zero. The adjacent difference is

$$\xi = \begin{cases} 0, & k < 0, \\ -d_0, & k = 0, \\ d_{k-1} \frac{k - (m+1)p}{kq}, & 1 \leq k \leq m+1, \\ 0, & k \geq m+2. \end{cases} \quad (2.20)$$

If $\xi \leq 0$, the quantity subtracted from $5V\xi$ in (2.19) is strictly positive. Indeed $V \geq x(1-x) + pq$ implies

$$\max\{V, S_2\} \geq \frac{1}{3},$$

by the same elementary calculation used in the proof of Lemma 2.3, and hence

$$C_E \left(8\sqrt{V} + \frac{2S_2}{\sqrt{V}} \right) + z\phi(z)(2 + z^2) \geq \frac{8C_E}{\sqrt{3}} - \frac{4}{e\sqrt{\pi}} > 0.$$

The right side of (2.19) is therefore negative. Thus only the positive range in the third line of (2.20) remains.

Put $y = k - (m+1)p > 0$, $f = y/(kq)$, and $j = k - 1$. Then $0 < f \leq 1$ and the binomial recurrences give

$$d_{j+1} = (1-f)d_j, \quad d_{j-1} = r_0 d_j, \quad r_0 = \frac{(k-1)q}{(m-k+2)p}.$$

The mass of these three atoms is at most one, so

$$\xi = f d_j \leq \frac{f}{2 - f + r_0}. \quad (2.21)$$

Eliminating m and k gives

$$r_0 = \frac{A_0}{B_0}, \quad A_0 = W - (1-f)q^2, \quad B_0 = p^2 + (1-f)(W + 2pq), \quad (2.22)$$

as well as

$$y = \frac{f(W + pq)}{1 - qf}, \quad C_E \left(8\sqrt{V} + \frac{2S_2}{\sqrt{V}} \right) = \frac{2C_E}{\sqrt{V}} \{4W + 2 + t_x t_p\}. \quad (2.23)$$

Finally, the normal term obeys the lower bound (2.17). Here $W = mpq \geq pq$, and (2.22) gives $A_0 \geq 0$. For $W \leq 3$, Lemma 2.3, together with (2.21)–(2.23), makes the right side of (2.19) strictly negative, contradicting local minimality.

For $W \geq 3$, Fourier inversion gives the analytic bound

$$|d_{k-1} - d_k| \leq \frac{4}{\pi} \int_0^{\pi/2} \sin u e^{-2W \sin^2 u} du < \frac{2}{\pi W}. \quad (2.24)$$

The last inequality follows on setting $s = \cos u$ and using $1 - s^2 \geq 1 - s$. Combining (2.24) with the curvature formula (2.19), and using $W \leq V \leq W + 1/2$, the negative of the right side of (2.19) is at least

$$8C_E \sqrt{V} - \frac{4}{e\sqrt{\pi}} - \frac{10V}{\pi W}.$$

This expression is concave in V , so its minimum is at $V = W$ or $V = W + 1/2$. Both endpoint functions increase with W . At $W = 3$, the rational bounds $C_E > 2/5$, $\sqrt{3} > 5/3$, $\sqrt{7/2} > 11/6$, $\pi > 3$, and $4/(e\sqrt{\pi}) < 1$ give the respective lower bounds 1 and $44/45$. This excludes $M \geq 2$.

When $M = 1$, Theorem 1.2, proved independently in Section 3, gives the required bound. Its proof uses the support endpoint estimates and two direct compact inequalities, and does not use the common-diameter theorem. This excludes the assumed negative closed deficit in the remaining case. Reflection (2.4) supplies the left CDF limit. $\square$

Proof of Theorem 1.1. Delete deterministic coordinates and scale h to one. Each remaining summand can then be written $X_i = B_i - p_i$, so the data in (2.1) are exactly the variance and sum of absolute third moments of the sum. If the bound in Theorem 1.1 failed, one of the deficits (2.2)–(2.3) would be negative at an integer threshold. Proposition 2.2 would produce a negative minimum in the family (2.9), contrary to Proposition 2.4. Esseen's i.i.d. Bernoulli sequence has equal support diameters and its ratio tends to C_E , so no smaller constant is valid in this class. $\square$

3 Two summands with arbitrary diameters

For independent Bernoulli variables B_i with $\mathbb{P}(B_i = 1) = p_i \in (0, 1)$, write $q_i = 1 - p_i$ and

$$X_i = h_i(B_i - p_i), \quad a_i = p_i q_i, \quad r_i = p_i^2 + q_i^2, \quad V = \sum_i a_i h_i^2 = B^2, \quad R = \sum_i a_i r_i h_i^3 = B^3 \ell.$$

We first record the estimate used when a compact calculation reaches a small variance.

Lemma 3.1 (Optimization over the diameters). *Let $0 < p_i < 1$ and $h_i \geq 0$, with $\sum_i a_i h_i^2 > 0$. Then*

$$\frac{\sum_i a_i r_i h_i^3}{(\sum_i a_i h_i^2)^{3/2}} \geq \left(\sum_i \frac{a_i}{r_i^2} \right)^{-1/2}. \quad (3.1)$$

Equality holds precisely when $r_i h_i$ is constant in i .

Proof. Hölder's inequality gives

$$\sum_i a_i h_i^2 = \sum_i (a_i r_i h_i^3)^{2/3} (a_i / r_i^2)^{1/3} \leq \left(\sum_i a_i r_i h_i^3 \right)^{2/3} \left(\sum_i a_i / r_i^2 \right)^{1/3}.$$

Its equality condition is $a_i r_i h_i^3$ proportional to a_i / r_i^2 . $\square$

The endpoint estimate below holds for any finite number of summands. It is also used in Section 6.

Proposition 3.2 (Support endpoints). *For every finite Bernoulli array with $0 < p_i < 1$ and $h_i > 0$, both $H_S^+(y)$ and $H_S^-(y)$ are strictly positive at each endpoint of the support of $S = \sum_i h_i(B_i - p_i)$.*

The proof is in Appendix A. It reduces the two CDF values containing an endpoint atom to an explicit rational inequality for the Bernoulli odds. The other two values follow from an elementary normal-tail estimate.

The interior atoms of a two-summand law are covered by the following two compact inequalities.

Lemma 3.3 (Two compact inequalities). *Let $0 < p_1, p_2 < 1$, $q_i = 1 - p_i$, $a_i = p_i q_i$, and $r_i = p_i^2 + q_i^2$. For $0 \leq t \leq 1$, put $V = a_1 t^2 + a_2$ and $R = a_1 r_1 t^3 + a_2 r_2$. Then*

$$C_E \frac{R}{V^{3/2}} + \Phi\left(\frac{-p_1 t + q_2}{\sqrt{V}}\right) - (1 - p_1 p_2) > 0. \quad (3.2)$$

For $0 \leq \theta \leq \pi/2$, set $\alpha = \cos \theta$, $\beta = \sin \theta$. Then

$$C_E \left(\alpha^3 \frac{r_1}{\sqrt{a_1}} + \beta^3 \frac{r_2}{\sqrt{a_2}} \right) + \Phi\left(\alpha \sqrt{\frac{q_1}{p_1}} - \beta \sqrt{\frac{p_2}{q_2}}\right) - q_2 > 0. \quad (3.3)$$

The proof, rational covers, and bounds at singular parameter faces are given in Section 7. Inequality (3.3) is proved for every $\alpha, \beta \geq 0$ with $\alpha^2 + \beta^2 = 1$. The parameters obtained from ordered diameters $0 < h_1 \leq h_2$ form a subset of this domain.

Proof of Theorem 1.2. Delete any deterministic coordinate. A remaining single Bernoulli variable is covered by Proposition 3.2. Otherwise relabel and scale so $h_2 = 1$ and $0 < h_1 = t \leq 1$. When $t < 1$, the four configurations have the order $00 < 10 < 01 < 11$. If $x_A = \sum_i (A_i - p_i) h_i$, their CDF values are

A	$F_S^-(x_A)$	$F_S^+(x_A)$
00	0	$q_1 q_2$
10	$q_1 q_2$	q_2
01	q_2	$1 - p_1 p_2$
11	$1 - p_1 p_2$	1.

Proposition 3.2 handles both endpoint atoms. The closed value at 01 is (3.2). At 10, put

$$\alpha = \frac{h_1 \sqrt{a_1}}{\sqrt{V}}, \quad \beta = \frac{h_2 \sqrt{a_2}}{\sqrt{V}}; \quad \alpha^2 + \beta^2 = 1.$$

Its closed deficit becomes (3.3). At $t = 1$ the two middle atoms merge; their left and closed CDF values are $q_1 q_2$ and $1 - p_1 p_2$. Thus (3.2) is exactly the merged closed deficit. The split value q_2 is not used at this collision. Complementing both Bernoulli variables gives $-S$ with the same variance, third absolute moments, and gap ordering. The identity $F_S^-(y) = 1 - F_{-S}^+(-y)$ supplies every left limit from the closed values just proved. Equation (1.5) completes the proof. A zero diameter or a probability equal to zero or one is handled by deletion before standardization. If no variance remains, the hypotheses of the theorem are not satisfied. $\square$

Corollary 3.4 (Mixtures of the two Esseen orientations). *Fix $c > 0$ and numbers $0 \leq \eta_i \leq 1$. Suppose the independent variables X_i have laws*

$$\mathcal{L}(X_i) = (1 - \eta_i) \mathcal{L}(cZ) + \eta_i \mathcal{L}(-cZ), \quad Z \sim \mu_E.$$

Then $\Delta \leq C_E \ell$.

Proof. Represent each mixture by an independent choice of orientation, followed by an independent draw from that oriented law. Conditional on the orientations, every summand has variance c^2 , absolute third moment $c^3 b_E$, where $b_E = (p_E^2 + q_E^2)/\sqrt{p_E q_E}$, and common support diameter $c/\sqrt{p_E q_E}$. Theorem 1.1 bounds every conditional CDF by the same $C_E \ell$ relative to the same standard normal CDF. Averaging preserves that bound. For $0 < \eta_i < 1$ the summand has four support points. $\square$

4 Atoms and short intervals

We use the product-measure Lubell–Yamamoto–Meshalkin (LYM) inequality in the form stated by Yehuda and Yehudayoff [11, Theorem 2]; their Theorem 1 gives the antichain bound used for the atom comparison. The increasing chain with rank-conditioned Bernoulli marginals was constructed earlier by Broman, van de Brug, Kager, and Meester [4, Lemma 2.2]; see also [11, Lemma 3 and Proposition 5]. We give an inductive construction, derive the equality condition, and extend the argument to closed intervals.

Proposition 4.1 (Weighted antichains and equality in the atom bound). *Let $B_1, \dots, B_n$ be independent Bernoulli variables with parameters $p_i \in (0, 1)$, let $h_i > 0$, and put $N = \sum_i B_i$. For every antichain $\mathcal{A} \subseteq 2^{[n]}$ (no member properly contains another),*

$$\sum_{A \in \mathcal{A}} \frac{\mathbb{P}(\{i : B_i = 1\} = A)}{\mathbb{P}(N = |A|)} \leq 1. \quad (4.1)$$

Consequently,

$$\sup_y \mathbb{P}\left(\sum_i h_i B_i = y\right) \leq \max_k \mathbb{P}(N = k). \quad (4.2)$$

If the mass function of N has a unique mode $m \in \{1, \dots, n-1\}$, then equality in the atom comparison (4.2) forces $h_1 = \dots = h_n$.

Proof. Write $r_i = p_i/(1-p_i)$ and let π_k be the conditional law of the random subset $\{i : B_i = 1\}$ given $N = k$. Thus

$$\pi_k(A) = \frac{\prod_{i \in A} r_i}{e_k(r_1, \dots, r_n)} \quad (|A| = k),$$

where e_k is the elementary symmetric polynomial. We first construct a random maximal chain $A_0 \subset A_1 \subset \dots \subset A_n$ whose rank- k marginal is π_k .

Proceed by induction on n . Write $\pi_j^{(-n)}$ for the analogous rank- j law on $[n-1]$. Let $\alpha_k = \pi_k\{n \in A\}$, with $\alpha_0 = 0$ and $\alpha_n = 1$. For $1 \leq k \leq n-1$,

$$\frac{\alpha_k}{1 - \alpha_k} = r_n \frac{e_{k-1}(r_1, \dots, r_{n-1})}{e_k(r_1, \dots, r_{n-1})}.$$

Newton's inequalities make the ratio on the right nondecreasing in k , so $\alpha_k \leq \alpha_{k+1}$. Couple ranks k and $k+1$ in three parts. With probability α_k , both sets contain n and their intersections with $[n-1]$ use the inductive coupling from $\pi_{k-1}^{(-n)}$ to $\pi_k^{(-n)}$. With probability $1 - \alpha_{k+1}$, neither set contains n and the intersections use the coupling from $\pi_k^{(-n)}$ to $\pi_{k+1}^{(-n)}$. With the remaining probability $\alpha_{k+1} - \alpha_k$, sample $C \sim \pi_k^{(-n)}$ and use $C \subset C \cup \{n\}$. Zero-mass branches are omitted; thus no nonexistent rank is invoked when $k = 0$ or $k = n-1$. Separating the contain- n and exclude- n cases verifies the two marginals. Composing these adjacent kernels gives the required chain.

Every maximal chain meets an antichain at most once. Taking expectations of the number of intersections gives the weighted LYM inequality (4.1). For fixed y , the fiber

$$\mathcal{F}_y = \left\{ A : \sum_{i \in A} h_i = y \right\}$$

is an antichain because every h_i is positive. If $s_k = \mathbb{P}(N = k)$ and $q_k = \pi_k(\mathcal{F}_y)$, then

$$\mathbb{P} \left(\sum_i h_i B_i = y \right) = \sum_k s_k q_k \leq \left(\max_k s_k \right) \sum_k q_k \leq \max_k s_k,$$

which proves the atom comparison (4.2).

Suppose equality holds, let y attain the supremum, and let m be the unique interior mode. If $M = \max_k s_k$, then the identity

$$M - \mathbb{P} \left(\sum_i h_i B_i = y \right) = M \left(1 - \sum_k q_k \right) + \sum_k (M - s_k) q_k$$

forces $q_m = 1$. Since every m -subset has positive π_m -mass, every such subset lies in $\mathcal{F}_y$. Given $i \neq j$, choose an $(m-1)$ -subset C disjoint from $\{i, j\}$. Comparing $C \cup \{i\}$ and $C \cup \{j\}$ gives $h_i = h_j$. $\square$

Proposition 4.2 (An atom and third-moment inequality). *Let $B_1, \dots, B_n$ be independent Bernoulli variables with $p_i = \mathbb{P}(B_i = 1) \in (0, 1)$. Let*

$$X_i = h_i(B_i - p_i), \quad h_i > 0, \quad a_i = p_i(1 - p_i), \quad t_i = 1 - 2p_i,$$

and put

$$S = \sum_i X_i, \quad V = \sum_i a_i h_i^2, \quad \kappa = \sum_i a_i h_i^3 t_i, \quad \rho = \frac{1}{2} \sum_i a_i h_i^3 (1 + t_i^2).$$

Choose $\varepsilon \in \{-1, 1\}$ so that $\varepsilon \kappa = |\kappa|$; when $\kappa \neq 0$, this is $\varepsilon = \text{sgn } \kappa$. Set

$$g_i = 1 + K_E(t_i - \varepsilon t_*)^2, \quad A = \sum_i \frac{a_i}{g_i^2}.$$

For $\nu \geq 1$ and $c \in (0, 1]$, define

$$\mathcal{M}(\nu, c) := \sqrt{\frac{\pi \nu c}{2}} \frac{1}{\pi} \int_0^\pi (1 - c \sin^2(\theta/2))^{\nu/2} d\theta.$$

The scalar inequality

$$\mathcal{M}(\nu, 1 - t^2) \leq 1 + K_E(t - t_*)^2 \quad (\nu \geq 1, \ 0 \leq t < 1), \quad (4.3)$$

is proved in Appendix B. It implies the joint atom and third-moment bound

$$\sqrt{2\pi V} \frac{1}{2} \sup_x \mathbb{P}(S = x) + \frac{|\kappa|}{6V} \leq K_E \frac{\rho}{V}, \quad (4.4)$$

for every finite array of centered two-point laws.

Proof. The identities $K_E t_* = 1/6$ and $K_E(1 - t_*^2) = 1$ give

$$K_E \rho_i = a_i h_i^3 \left(\frac{1}{2} + \frac{\varepsilon t_i}{6} \right) + \frac{K_E a_i h_i^3}{2} (t_i - \varepsilon t_*)^2, \quad \rho_i = \frac{1}{2} a_i h_i^3 (1 + t_i^2).$$

Since $\varepsilon \kappa = |\kappa|$, the inequality (4.4) is equivalent to

$$\sup_x \mathbb{P}(S = x) \sqrt{2\pi V} \leq \frac{1}{V} \sum_i a_i g_i h_i^3. \quad (4.5)$$

Let $q_{\max} = \max_k \mathbb{P}(N = k)$, where $N = \sum_i B_i$. Translation and Proposition 4.1 give $\sup_x \mathbb{P}(S = x) \leq q_{\max}$. Hölder's inequality gives

$$V = \sum_i (a_i g_i h_i^3)^{2/3} (a_i / g_i^2)^{1/3} \leq \left(\sum_i a_i g_i h_i^3 \right)^{2/3} A^{1/3}.$$

Thus (4.5) follows if

$$q_{\max} \sqrt{2\pi A} \leq 1. \quad (4.6)$$

Equality in this application of Hölder's inequality holds when h_i is proportional to g_i^{-1} .

Put $c_i = 4a_i = 1 - t_i^2$, $w_i = a_i / (A g_i^2)$, and $\nu_i = w_i^{-1}$. Then $\sum_i w_i = 1$ and $\nu_i c_i = 4A g_i^2$. Fourier inversion and generalized Hölder give

$$q_{\max} \sqrt{2\pi A} \leq \prod_i \left\{ \frac{\mathcal{M}(\nu_i, c_i)}{g_i} \right\}^{w_i}.$$

If $\varepsilon t_i \geq 0$, the scalar condition (4.3) applied to $t = \varepsilon t_i$ bounds the corresponding factor by one. If $\varepsilon t_i < 0$, replace it by $|\varepsilon t_i|$; the left side is unchanged and

$$1 + K_E(t_i - \varepsilon t_*)^2 = 1 + K_E(|t_i| + t_*)^2 \geq 1 + K_E(|t_i| - t_*)^2.$$

Hence every factor is at most one, proving the modal bound (4.6). $\square$

For any atom y , put $m_y = \mathbb{P}(S = y)$ and $F_S^0(y) = \{F_S^+(y) + F_S^-(y)\}/2$. Then

$$\max\{F_S^+(y) - \Phi(y/B), \Phi(y/B) - F_S^-(y)\} = \frac{m_y}{2} + |F_S^0(y) - \Phi(y/B)|.$$

Dividing (4.4) by $\sqrt{2\pi V}$ gives

$$\frac{1}{2} \sup_y \mathbb{P}(S = y) + \frac{|\kappa|}{6\sqrt{2\pi} V^{3/2}} \leq C_E \ell.$$

The second term is the uniform amplitude of $\kappa(1 - z^2)\phi(z)/(6V^{3/2})$, since $\sup_z |(1 - z^2)\phi(z)| = \phi(0)$. A bound for the remaining midpoint CDF error at the same threshold is still needed for the bound with arbitrary diameters in Problem 6.1.

Theorem 4.3 (Closed intervals). *Let $Y = \sum_{i=1}^n h_i B_i$, where the B_i are independent, $0 < p_i < 1$, and $h_i > 0$. For $r \geq 0$, put $Q(Y, r) = \sup_x \mathbb{P}\{Y \in [x, x + r]\}$. Choose a nonempty $J \subseteq \{1, \dots, n\}$, put $d = |J|$, $\delta_J = \min_{i \in J} h_i$, and let $s_{(1)}^J \geq \dots \geq s_{(d+1)}^J$ be the decreasing arrangement of the rank probabilities of $N_J = \sum_{i \in J} B_i$. Then*

$$Q(Y, r) \leq \sum_{a=1}^L s_{(a)}^J, \quad L = \min\{\lfloor r/\delta_J \rfloor + 1, d + 1\}. \quad (4.7)$$

If $r < \delta_J$, N_J has a unique mode $m \in \{1, \dots, d - 1\}$, and equality holds, all selected m -subset sums lie in one interval of length r . Consequently $|h_i - h_j| \leq r$ for every $i, j \in J$.

Proof. Condition on the variables outside J . For each outside configuration the interval is translated, but its length remains r . Use the maximal chain $A_0 \subset \dots \subset A_d$ from Proposition 4.1 on J , and set $H_k = \sum_{i \in A_k} h_i$. Along each chain, $H_{k+1} - H_k \geq \delta_J$, so a closed interval of length r contains at most L chain values. If q_k is the probability that H_k lies in the translated interval, then $0 \leq q_k \leq 1$ and $\sum_k q_k \leq L$. The conditional interval probability is $\sum_k s_k^J q_k$. Choose a set T of L largest ranks and put $\lambda = \min_{k \in T} s_k^J$. The exact identity

$$\sum_{k \in T} s_k^J - \sum_k s_k^J q_k = \sum_{k \in T} (s_k^J - \lambda)(1 - q_k) + \sum_{k \notin T} (\lambda - s_k^J) q_k + \lambda \left(L - \sum_k q_k \right)$$

has nonnegative terms, also when the cutoff rank mass is tied. Averaging over the outside configurations proves (4.7).

For equality with $L = 1$, every positive-probability outside configuration must itself attain the same bound. Fix one such configuration. Uniqueness of the mode in the preceding identity forces $q_m = 1$ and $q_k = 0$ for $k \neq m$. Every m -subset has positive conditional rank mass, so all of its subset sums lie in this one translated interval. For distinct $i, j \in J$, choose an $(m-1)$ -subset C disjoint from them. The sums for $C \cup \{i\}$ and $C \cup \{j\}$ differ by $h_i - h_j$, proving the assertion. $\square$

The strict cutoff $r < \delta_J$ is necessary for closed intervals: a single Bernoulli variable with gap r has its entire mass in $[0, r]$. Uniqueness and interiority of the mode enter precisely in the equality argument. A boundary mode can attain the atom bound for unequal gaps.

Corollary 4.4 (Interval concentration and selected moments). *For $S = \sum_i h_i(B_i - p_i)$, let $J \neq \emptyset$ satisfy $h_i > r$ on J . Write*

$$V_J = \sum_{i \in J} p_i q_i h_i^2, \quad \kappa_J = \sum_{i \in J} p_i q_i (q_i - p_i) h_i^3, \quad R_J = \sum_{i \in J} p_i q_i (p_i^2 + q_i^2) h_i^3.$$

Then

$$\frac{1}{2} \sqrt{2\pi V_J} Q(S, r) + \frac{|\kappa_J|}{6V_J} \leq K_E \frac{R_J}{V_J}. \quad (4.8)$$

With $\vartheta_J = V_J/B^2$, the full-variance form is

$$\vartheta_J^{3/2} \frac{\sqrt{2\pi}}{2} Q(S, r) + \frac{|\kappa_J|}{6B^3} \leq K_E \frac{R_J}{B^3}. \quad (4.9)$$

Proof. Theorem 4.3 gives $Q(S, r) \leq q_J := \max_k \mathbb{P}(N_J = k)$. The proof of (4.4) establishes the stronger modal inequality with q_J in place of the atom mass of the selected weighted sum. Substitution proves (4.8); multiplication by V_J/B^3 proves (4.9). $\square$

Near equality in the rank bound (4.7) with $L = 1$ gives no uniform closeness of the diameters. For example, let $0 < \eta < 1/2$, $p_1 = \eta$, $p_2 = 1/2$, $h_1 = M > 1 + r$, $h_2 = 1$, and $0 \leq r < 1$. The unique modal rank is 1, of mass $1/2$, while $Q(Y, r) = \mathbb{P}\{Y \in [1, 1 + r]\} = (1 - \eta)/2$. Thus the difference from the rank bound is $\eta/2$, although $M - 1$ can be arbitrarily large.

5 Moment optimization and three-point reduction

At a fixed threshold, convolution is affine in each summand law. Fixing its mean and variance therefore gives a moment optimization problem with three affine constraints. Extreme-point theorems for moment sets, including Winkler's characterization [10, Theorem 2.1], provide the broader setting. Strong duality under interior moment constraints is also classical [3, Theorem 2.2]. We give the compactness and separation arguments for the precise coercive objectives needed here.

Theorem 5.1 (Strong quadratic moment duality). *Let $s > 0$ and let $h : \mathbb{R} \rightarrow \mathbb{R}$ be upper semicontinuous with*

$$-K_-(1 + |u|^3) \leq h(u) \leq K_+ - \lambda|u|^3$$

for some finite K_- , K_+ and $\lambda > 0$. Then

$$\max_{P: \int dP=1, \int u dP=0, \int u^2 dP=s, \int |u|^3 dP < \infty} \int h dP = \min_{\substack{q(u)=A+Bu+Cu^2 \\ q \geq h \text{ on } \mathbb{R}}} (A + Cs). \quad (5.1)$$

Both extrema are attained. An extreme primal optimizer has at most three atoms, and every primal optimizer is concentrated on the contact set of every optimal quadratic majorant.

Proof. Fix any finite-support feasible law P_0 . A maximizing sequence may be taken with value at least $\int h dP_0 - 1$; together with $h \leq K_+ - \lambda|u|^3$, this gives uniformly bounded third moments. The laws are tight, and the uniform bound on third moments makes u^2 uniformly integrable. A weak limit preserves the three moment constraints and has finite third moment by Fatou's lemma. The bounded-above upper semicontinuous version of Portmanteau's theorem shows that the limit attains the maximum. The maximizing set is weakly compact and convex; it is a face of the feasible set because the objective is affine. It has an extreme point, which is also extreme in the feasible set.

If an extreme feasible law P had four disjoint Borel sets $A_1, \dots, A_4$ of positive mass, their vectors

$$\left(P(A_j), \int_{A_j} u dP, \int_{A_j} u^2 dP \right) \in \mathbb{R}^3$$

would be linearly dependent. A nonzero bounded function $g = \sum_j c_j \mathbf{1}_{A_j}$ would then satisfy $\int g dP = \int u g dP = \int u^2 g dP = 0$. For sufficiently small $\epsilon > 0$, the two distinct laws $(1 \pm \epsilon g)P$ remain feasible and have finite third moments, while their average is P . Thus an extreme law has at most three support points.

For strong duality, let $\mathcal{D} \subset \mathbb{R}^4$ consist of

$$\left(\mu(\mathbb{R}), \int u d\mu, \int u^2 d\mu, z \right), \quad \mu \geq 0, \quad z \leq \int h d\mu,$$

where μ has finite third moment. This is a convex cone. It is closed: along a convergent sequence with z bounded below, the inequalities $z \leq \int h d\mu \leq K_+ \mu(\mathbb{R}) - \lambda \int |u|^3 d\mu$ give a uniform third-moment bound; weak compactness, uniform integrability of u^2 , and Portmanteau for the bounded-above upper semicontinuous function h then retain all four inequalities in the limit.

Let M denote the primal maximum. The point $p = (1, 0, s, M)$ belongs to $\mathcal{D}$, whereas $p + (0, 0, 0, \epsilon)$ does not for any $\epsilon > 0$, so p is a boundary point. A nonzero supporting functional (a, b, c, d) exists; since $\mathcal{D}$ is a cone and contains both 0 and $2p$, its supporting level at p is zero. Thus $am_0 + bm_1 + cm_2 + dz \leq 0$ on $\mathcal{D}$ and equality holds at p . The downward vertical ray in $\mathcal{D}$ gives $d \geq 0$. In fact $d > 0$: if $d = 0$, then (a, b, c) would support the moment cone

$$\{0\} \cup \{(m_0, m_1, m_2) : m_0 > 0, m_2 \geq m_1^2/m_0\}$$

at $(1, 0, s)$, which is an interior point because $s > 0$, forcing the whole functional to vanish. Normalize $d = 1$. Applying the support inequality to a point mass at u gives $a + bu + cu^2 + h(u) \leq 0$, so $q(u) := -a - bu - cu^2$ majorizes h . Equality at p gives $M = -a - cs = A + Cs$. For every feasible P and every feasible q , integration gives $\int h dP \leq A + Cs$. Thus this attained dual value is minimal. Finally, for any primal and dual optimizers, $0 = (A + Cs) - \int h dP = \int (q - h) dP$ with $q - h \geq 0$, proving the contact assertion. $\square$

Theorem 5.2 (Three-point reduction). *Every strict failure of $\Delta \leq C_E \ell$ can be replaced by a strict failure with the same number of independent centered summands, each supported on at most three points.*

Proof. Fix a strict one-sided failure, its threshold y , its choice $\varepsilon \in \{+1, -1\}$, and the variances of all summands. Put $\lambda = C_E B^{-3} > 0$. When all coordinates except X_i are fixed, let $T_i = \sum_{j \neq i} X_j$ and set

$$h_i(u) = \varepsilon F_{T_i}^\varepsilon(y - u) - \lambda |u|^3.$$

The first term is upper semicontinuous for both signs. Indeed the closed CDF composed with $y - u$ is upper semicontinuous, while the strict CDF composed with $y - u$ is lower semicontinuous. The cubic term gives the two bounds required in Theorem 5.1. A coordinate of zero variance is deterministic. Every other coordinate can be replaced by an extreme maximizer with its mean and variance unchanged. This increases or preserves

$$\varepsilon F_{\sum_i X_i}^\varepsilon(y) - \varepsilon \Phi(y/B) - \lambda \sum_i \mathbb{E}|X_i|^3.$$

Sequential replacement gives the claimed finite-support failure. $\square$

6 The remaining finite problems

We formulate two open inequalities whose simultaneous proof would imply the unrestricted sharp bound. The first is the bound for Bernoulli arrays with arbitrary positive diameters. The second concerns the centered three-point laws supplied by Theorem 5.2.

6.1 A minimal counterexample in the two-point class

For $n \geq 1$, $p \in (0, 1)^n$, and $h \in (0, \infty)^n$, put

$$a_i = p_i q_i, \quad r_i = p_i^2 + q_i^2, \quad \sum_i a_i h_i^2 = 1, \quad R = \sum_i a_i r_i h_i^3.$$

For $A \in \{0, 1\}^n$, set

$$x_A = \sum_i (A_i - p_i) h_i, \quad D_A(p, h) = C_E R + \Phi(x_A) - \sum_{C \in \{0, 1\}^n} \pi_C(p) \mathbf{1}_{\{(C-A) \cdot h \leq 0\}}, \quad (6.1)$$

where $\pi_C(p) = \prod_i p_i^{C_i} q_i^{1-C_i}$.

Problem 6.1 (Arbitrary Bernoulli diameters). Prove $D_A(p, h) \geq 0$ on the complete domain in (6.1), for every n and A .

Equation (1.5), reflection, and scaling show that this problem is equivalent to the sharp bound for all finite arrays of two-point laws. The following attainment statement removes the zero-probability and infinite-diameter boundaries from a least-dimensional failure.

Theorem 6.2 (Least-dimensional two-point failure). *If Problem 6.1 fails, let n be its least failing dimension and let $d_n < 0$ be the infimum of (6.1) in that dimension. Then d_n is attained at probabilities in $(0, 1)$ and finite positive diameters, with every variance share $p_i q_i h_i^2 > 0$. One has $n \geq 3$, the diameters are not all equal, and the selected threshold is strictly between the two support endpoints. Subset-sum collisions may occur.*

Proof. Since $D_A \geq C_E R - 1$, the infimum is finite and every negative minimizing sequence has uniformly bounded R . Put $w_i = a_i h_i^2$, so $\sum_i w_i = 1$. The contribution to R is $r_i w_i^{3/2} / \sqrt{a_i}$, and $r_i \geq 1/2$ gives

$$w_i^3 / a_i \leq 4R^2.$$

Pass to a subsequence on which A is fixed, $p_i \rightarrow p_i^*$, and $w_i \rightarrow w_i^*$. The set $L = \{i : w_i^* > 0\}$ is nonempty. For $i \in L$, the last inequality gives $0 < p_i^* < 1$, and $h_i \rightarrow \sqrt{w_i^* / (p_i^* q_i^*)} \in (0, \infty)$. The centered sum of coordinates outside L tends to zero in L^2 . Thus the full sums converge weakly to the limiting sum S_L , which still has variance one.

Pass to a further subsequence for which x_A converges in the extended real line. If its limit is $+\infty$ or $-\infty$, tightness and the normal tail give a nonnegative lower limit for the deficit, a contradiction. If $x_A \rightarrow z \in \mathbb{R}$, moving-threshold Portmanteau gives

$$\limsup \mathbb{P}(S \leq x_A) \leq \mathbb{P}(S_L \leq z).$$

The moments of the retained coordinates converge; those of the discarded coordinates are nonnegative. Therefore

$$d_n \geq C_E R_L + \Phi(z) - \mathbb{P}(S_L \leq z).$$

The right side is negative. Its CDF is then nonzero, so moving z left to the greatest retained support point below it can only decrease that deficit. Minimality of n forces $L = \{1, \dots, n\}$.

At this interior limit, each term $\pi_C(p) \mathbf{1}_{\{(C-A) \cdot h \leq 0\}}$ is upper semicontinuous. The other terms in (6.1) are continuous. Consequently D_A is lower semicontinuous and its limit value is at most d_n . By the definition of the infimum it equals d_n . Theorem 1.2 gives $n \geq 3$; Theorem 1.1 excludes equal diameters; and Proposition 3.2 excludes support endpoints. $\square$

At the attained point, variations that preserve all subset-sum collisions satisfy the following necessary conditions. Let M have independent rows spanning every subset-sum equality active at h , so $Mh = 0$. All other subset orders are strict in a relative neighborhood of h . Set $d_i = A_i - p_i$, $t_i = 1 - 2p_i$, $x = x_A$, and $y = \sum_i A_i h_i$. There are multipliers λ and γ such that

$$3C_E a_i r_i h_i^2 + \phi(x) d_i - 2\lambda a_i h_i + (M^\top \gamma)_i = 0, \quad (6.2)$$

$$C_E t_i^3 h_i^3 - \phi(x) h_i + I_i - \lambda t_i h_i^2 = 0, \quad I_i = \mathbb{P}\{y - h_i < Y_{-i} \leq y\}, \quad (6.3)$$

$$2\lambda = 3C_E R + x\phi(x), \quad (6.4)$$

where $Y_{-i} = \sum_{j \neq i} h_j B_j$. Moreover, for every v with $Mv = 0$ and $\sum_i a_i h_i v_i = 0$,

$$\sum_i (6C_E a_i r_i h_i - 2\lambda a_i) v_i^2 - x\phi(x) \left(\sum_i d_i v_i \right)^2 \geq 0. \quad (6.5)$$

To verify these conditions, apply the Lagrange multiplier rule to $D_A - \lambda(\sum_i a_i h_i^2 - 1) + \gamma \cdot Mh$ on the fixed stratum. The constraint qualification holds because $Mh = 0$, whereas the variance gradient has scalar product 2 with h . Differentiation uses $(a_i r_i)' = t_i^3$, $a_i' = t_i$, and $\partial_{p_i} F_S^+(x_A) = -I_i$. Multiplication of (6.2) by h_i and summation gives (6.4); the gap Hessian is (6.5). Changing the representative A of a colliding atom adds a vector in the row span of M . Its contribution is absorbed by γ , and its scalar product with every allowed v is zero. A proof that no negative point satisfies this system on any complete collision stratum would solve Problem 6.1.

6.2 An open inequality for three-point mixtures

For $a, b > 0$, define the centered pair

$$Q_{a,b} = \frac{b}{a+b}\delta_{-a} + \frac{a}{a+b}\delta_b, \quad \text{Var}(Q_{a,b}) = ab, \quad \int |u|^3 dQ_{a,b}(u) = \frac{ab(a^2 + b^2)}{a+b}.$$

A centered law with exactly three support points has one of the representations

$$P = \alpha\delta_0 + (1-\alpha)Q_{a,b}, \quad \text{or, up to reflection,} \quad P = \alpha Q_{a,b_1} + (1-\alpha)Q_{a,b_2}, \quad 0 < b_1 < b_2, \quad (6.6)$$

with $0 < \alpha < 1$. The first formula follows from centering on $\{-a, 0, b\}$. For masses m_-, m_1, m_2 at $-a, b_1, b_2$, centering gives $am_- = b_1m_1 + b_2m_2$; the mixture weights are $m_j(a + b_j)/a$ and sum to one. If P is an extreme optimizer in Theorem 5.1, every point in (6.6) is a contact of the same optimal quadratic. Each component law therefore maximizes the same coordinate objective among centered laws of its own variance. Indeed, at variance r the quadratic bounds the objective by $A + Cr$, and a component supported on contacts attains this value.

Fix $T = \sum_j T_j$, where the T_j are independent centered variables with at most three atoms each, and put $V_T = \text{Var}(T) > 0$ and $R_T = \sum_j \mathbb{E}|T_j|^3$. Choose $\varepsilon \in \{+1, -1\}$ and $y \in \mathbb{R}$. Let $P = \alpha Q_- + (1-\alpha)Q_+$ be one of (6.6), and write

$$r_- = \text{Var}(Q_-) < r_+ = \text{Var}(Q_+), \quad s = \alpha r_- + (1-\alpha)r_+, \quad V = V_T + s, \quad \lambda = C_E V^{-3/2}.$$

Suppose P is an extreme maximizing law, of variance s , for $h(u) = \varepsilon F_T^\varepsilon(y - u) - \lambda|u|^3$. For independent $X_\pm \sim Q_\pm$, define

$$\begin{aligned} V_\pm &= V_T + r_\pm, \quad R_\pm = R_T + \mathbb{E}|X_\pm|^3, \quad R = \alpha R_- + (1-\alpha)R_+, \\ \delta_\pm &= \varepsilon \Phi(y/\sqrt{V_\pm}) + C_E R_\pm V_\pm^{-3/2} - \varepsilon F_{T+X_\pm}^\varepsilon(y). \end{aligned} \quad (6.7)$$

With $X \sim P$ independent of T , the exact excess of the mixture is

$$\varepsilon F_{T+X}^\varepsilon(y) - \varepsilon \Phi(y/\sqrt{V}) - C_E R V^{-3/2} = G - \alpha \delta_- - (1-\alpha)\delta_+, \quad (6.8)$$

where

$$\begin{aligned} G &= \varepsilon \left\{ \alpha \Phi(y/\sqrt{V_-}) + (1-\alpha) \Phi(y/\sqrt{V_+}) - \Phi(y/\sqrt{V}) \right\} \\ &\quad + C_E \left\{ \alpha R_- V_-^{-3/2} + (1-\alpha) R_+ V_+^{-3/2} - R V^{-3/2} \right\}. \end{aligned} \quad (6.9)$$

This follows by adding and subtracting the endpoint normal approximations; both convolution and the third absolute moment are affine in the mixture.

Problem 6.3 (Three-point mixture). In the complete domain just specified, suppose that both endpoint arrays $T + X_-$ and $T + X_+$ satisfy the sharp Berry–Esseen bound at their respective variances. Prove

$$\alpha \delta_- + (1-\alpha)\delta_+ \geq G. \quad (6.10)$$

Proposition 6.4 (Reduction to the two finite problems). A solution of Problems 6.1 and 6.3 implies $\Delta \leq C_{El}$ for every independent array in (1.1). If the two-point problem is solved but the general bound fails, Problem 6.3 has a counterexample with $G > 0$.

Proof. Use Theorem 5.2 and choose a strict counterexample with the fewest coordinates having exactly three atoms. Under Problem 6.1, this number is positive. The one-summand estimate of Bentkus and Kirsas [2, Theorem 1] gives $\Delta \leq c_{\text{BK}}\ell$ with $c_{\text{BK}} = 0.370352\dots < C_E$ for every centered law with finite absolute third moment. It excludes a single nondegenerate coordinate; hence deleting any chosen three-point coordinate leaves positive variance. Optimize that coordinate at its original variance and the selected sign and threshold. An extreme optimizer exists by Theorem 5.1. It cannot have two atoms, since that would give a strict failure with fewer three-point coordinates. Thus it has exactly three atoms and admits (6.6). Each endpoint replacement removes one three-point coordinate. By minimality, each endpoint array satisfies the sharp bound, so $\delta_-, \delta_+ \geq 0$ at the selected threshold. The optimized mixture still has positive excess. Equation (6.8) now gives $G > \alpha\delta_- + (1 - \alpha)\delta_+ \geq 0$, contradicting (6.10). $\square$

7 Computational proof supplement

Five primary calculations are used in the proofs: the exceptional-Bernoulli curvature inequality (2.15), the two inequalities of Lemma 3.3, the rational odds comparison in Appendix A, and the scalar Fourier comparison in Appendix B. Their mathematical domains and analytic complements are part of the proofs. The programs under `verify/finite/` implement the finite parts. The reported primary results and source hashes are in `certificates/finite/results.json`; `certificates/finite/README.md` lists the programs and rerun commands. Separate evaluators are supplied for the scalar atom inequality, the odds comparison, and the first interior two-summand threshold. The second threshold has a higher-precision rerun and separate formula and boundary checks. Paths are relative to the public supplement root.

The interval calculations use Arb through `python-flint` 0.8.0. Arb encloses real quantities by balls with certified outward rounding, including the elementary and special functions used here [7]. A numerical estimate without an enclosure is never an acceptance test. Every rational input box is enclosed; a lower bound is accepted only if its complete outward interval is positive. The use of floating-point scores to choose a subdivision direction in one program cannot change this test or the coverage identity.

7.1 The two-summand covers

For (3.2), `verify/finite/two_summands/certificate.py` starts from $[0, 1]^3$ in (p_1, p_2, t) and uses 192-bit arithmetic. For a probability interval $[l, u]$, its variance range has lower endpoint $\min\{l(1-l), u(1-u)\}$ and upper endpoint $1/4$ if the interval contains $1/2$, otherwise $\max\{l(1-l), u(1-u)\}$. The program obtains a lower bound for $R/V^{3/2}$ by taking the greater of direct interval evaluation, when V is bounded away from zero, and (3.1). In the latter bound it uses the upper enclosure of $a_1/r_1^2 + a_2/r_2^2$ before taking its reciprocal square root. It subtracts an upper enclosure of $1 - p_1p_2$.

For the normal term, direct evaluation applies when the variance interval has positive lower endpoint. If the variance interval meets zero and the numerator has nonnegative lower endpoint, dividing that endpoint by the upper variance square root gives a valid lower bound for the argument. A sign-indefinite numerator receives the lower CDF bound zero. The additional inequality

$$\frac{-p_1t + q_2}{\sqrt{a_1t^2 + a_2}} \geq -\sqrt{p_1/q_1}$$

provides a second bound. If the numerator is negative, its magnitude is at most p_1t and the denominator is at least $\sqrt{a_1}t$; otherwise the claim is immediate. The analogous bound $(q_1t - p_2)/\sqrt{V} \geq -\sqrt{p_2/q_2}$ is used by the accompanying checks. On boxes touching the corresponding

deterministic probability, the limiting CDF lower bound zero avoids a singular odds evaluation. Thus every bound remains valid on the positive-variance part of a box which touches a boundary face.

A box failing the strict positivity test is bisected into two closed rational boxes along one coordinate. The adaptive choice of coordinate uses lower bounds only as a heuristic; both children are retained. The complete calculation processed 2231 boxes and accepted 1116, with maximum accepted depth 16. The smallest accepted outward lower endpoint exceeded $3.2325447770 \times 10^{-5}$.

For (3.3), `verify/finite/two_summands/row10_certificate.py` starts from $[0, 1]^3$ in $(p_1, p_2, 2\theta/\pi)$ and uses 192-bit arithmetic. The shape

$$s(p) = \frac{p^2 + (1-p)^2}{\sqrt{p(1-p)}}$$

is decreasing on $(0, 1/2]$ and increasing on $[1/2, 1)$. For a probability interval its infimum is therefore 1 if the interval contains $1/2$, and otherwise its value at the endpoint nearest $1/2$. The two nonnegative third-moment terms are bounded separately using lower bounds for $\cos \theta$, $\sin \theta$, and $s(p_i)$. For the argument of Φ , the positive term uses the lower bound for $\cos \theta$ and the largest p_1 ; the negative term uses the upper bound for $\sin \theta$ and the largest p_2 . If the latter endpoint is 1, the CDF lower bound is zero. If the former endpoint is 1, its positive term is replaced by zero. This treats all boxes meeting probability boundaries without assigning values to an undefined standardized law. The upper bound for the subtracted q_2 uses the smallest p_2 . The wider coordinate is bisected whenever the bound is inconclusive. The complete cover processed 3509 boxes and accepted 1755, with maximum depth 17 and smallest outward lower endpoint greater than $1.3393433322 \times 10^{-5}$.

Each run starts with one box and replaces every unaccepted box by both children. Its terminal queue is empty, and its counts satisfy $N_{\text{processed}} = 2N_{\text{accepted}} - 1$. Depth or box-budget exhaustion terminates with an error rather than a positive conclusion. These facts establish coverage of the whole cubes. The first algebraic inequality extends to every positive-variance cube point. Its interpretation as the middle closed CDF requires $t > 0$. At $t = 0$ the correct distribution is obtained by deleting the zero-diameter coordinate. Probability endpoints are likewise handled by deletion in the proof of Theorem 1.2.

7.2 The other finite calculations

The exceptional-Bernoulli calculation uses three four-dimensional unit-box charts displayed in the proof of Lemma 2.3. It starts with one box per chart; the recorded complete cover has 448835 processed boxes and 224419 accepted boxes, satisfying $N_{\text{processed}} = 2N_{\text{accepted}} - 3$. Its maximum accepted depth is 31, and the smallest accepted outward lower endpoint exceeds $1.8532764256 \times 10^{-7}$. The separate analytic inequalities in that proof cover small total variance, binomial variance $W \geq 3$, and nonpositive adjacent mass differences. The direct two-summand proof above supplies the remaining case $M = 1$.

The scalar atom calculation is a fixed cover of 63 order slabs by 2048 intervals each. Formula (B.4) bounds every real order in each slab; a grid of real orders is not used. The positive polynomial (B.3) is evaluated in Arb, and all 129024 boxes satisfy the strict bound (B.5). An independently written evaluator uses direct coefficient sums instead of Horner evaluation. The infinite-order complement is reduced analytically to the single Bessel inequality stated in Appendix B and included in the verification material.

The odds comparison uses SymPy [8] for exact arithmetic in $\mathbb{Q}(\sqrt{5})$. For a polynomial $P(r, t) =$

$\sum c_{ij} r^i t^j$ and degrees D, E , its tensor Bernstein coefficients on $[0, 1]^2$ are

$$b_{kl} = \sum_{i \leq k, j \leq l} c_{ij} \frac{\binom{k}{i}}{\binom{D}{i}} \frac{\binom{l}{j}}{\binom{E}{j}}. \quad (7.1)$$

The identity $P = \sum_{k,l} b_{kl} \binom{D}{k} r^k (1-r)^{D-k} \binom{E}{l} t^l (1-t)^{E-l}$ turns coefficient signs into signs on the whole square. All signs in $\mathbb{Q}(\sqrt{5})$ can be determined by rational arithmetic, squaring only after the signs of the rational parts are known. Appendix A specifies the two mapped polynomials, the degrees, and every zero coefficient. The independent checker cancels a positive polynomial factor before a second basis conversion.

A Support endpoint estimates

Use the Bernoulli representation from Section 3. In this appendix $V = B^2$ and $\mathcal{B}_h = B^3 \ell$.

Lemma A.1 (Normal tails at the support endpoints). *Let $n \geq 1$, and let $B_1, \dots, B_n$ be independent, with $B_i \sim \text{Bern}(p_i)$, $p_i \in (0, 1)$, and $h_i > 0$. Put $q_i = 1 - p_i$ and define*

$$V = \sum_i p_i q_i h_i^2, \quad \mathcal{B}_h = \sum_i p_i q_i (p_i^2 + q_i^2) h_i^3,$$

and set

$$u = \sum_i q_i h_i, \quad m = \sum_i p_i h_i.$$

Then

$$C_E \frac{\mathcal{B}_h}{V^{3/2}} > \max \left\{ 1 - \Phi \left(\frac{u}{\sqrt{V}} \right), \Phi \left(-\frac{m}{\sqrt{V}} \right) \right\}. \quad (\text{A.1})$$

Consequently, the closed deficit at the upper support and the strict deficit at the lower support are both strictly positive.

Proof. It is enough to prove the first inequality in (A.1); complementing every B_i exchanges p_i and q_i , preserves V and $\mathcal{B}_h$, and gives the second. Normalize $V = 1$ and write

$$x_i = q_i h_i, \quad v_i = p_i q_i h_i^2, \quad b_i = p_i q_i (p_i^2 + q_i^2) h_i^3, \quad x = \sum_i x_i.$$

For $c = 2(\sqrt{2} - 1)$,

$$b_i = \frac{p_i^2 + q_i^2}{p_i} \frac{v_i^2}{x_i}, \quad \frac{p_i^2 + q_i^2}{p_i} - c = \frac{(\sqrt{2} p_i - 1)^2}{p_i} \geq 0.$$

Since $\sum_i v_i = 1$, Cauchy–Schwarz gives

$$\mathcal{B}_h \geq c \sum_i \frac{v_i^2}{x_i} \geq \frac{c}{x}. \quad (\text{A.2})$$

We claim that

$$x \{1 - \Phi(x)\} < C_E c \quad (x > 0). \quad (\text{A.3})$$

If $0 < x \leq 2/3$, then the left side is at most $1/3$. The elementary bounds

$$\sqrt{10} > \frac{79}{25}, \quad \sqrt{2} > \frac{169}{120}, \quad \pi < \frac{22}{7},$$

give $C_E > 49/120$ and $c > 49/60$. For the first comparison,

$$20(\sqrt{10} + 3) > \frac{616}{5} > 49\sqrt{\frac{44}{7}} > 49\sqrt{2\pi};$$

the middle inequality follows on squaring, with difference $2156/25$. The bound on π follows, for example, from

$$\frac{22}{7} - \pi = \int_0^1 \frac{t^4(1-t)^4}{1+t^2} dt > 0.$$

Hence $C_E c > 2401/7200 > 1/3$. For $x \geq 2/3$, Mills' inequality and monotonicity of ϕ give

$$x\{1 - \Phi(x)\} < \phi(x) \leq \phi(2/3) = \phi(0)e^{-2/9}.$$

Now $e^{-2/9} < 9/11 < c$, where the last inequality follows from $\sqrt{2} > 31/22$, and $C_E/\phi(0) = (\sqrt{10} + 3)/6 > 1$. This proves (A.3). Combining it with (A.2) proves the upper-support inequality. At that support the closed CDF equals one. At the lower support the strict CDF equals zero, and exact complementation exchanges the two CDF limits. The two deficit statements follow. $\square$

Proposition A.2 (An inequality for Bernoulli odds). *Let $o_1, \dots, o_n \geq 0$, not all zero, and put*

$$s = \sum_i o_i, \quad \Pi = \prod_i (1 + o_i)^{-1}, \quad g(o) = \frac{o(1+o)^2}{(1+o^2)^2}, \quad \mathcal{A} = \sum_i g(o_i). \quad (\text{A.4})$$

Then

$$\Pi^2 \mathcal{A} \leq \frac{1}{3} \quad (0 < s \leq 1), \quad (\text{A.5})$$

$$\Pi^2 \mathcal{A} \leq \frac{s}{(1+s/2)^2(1+s^2/4)^2} \quad \left(1 \leq s \leq \frac{16}{9}\right), \quad (\text{A.6})$$

and

$$\Pi^2 \mathcal{A} \leq \frac{944784}{6076225} \quad \left(s \geq \frac{16}{9}\right). \quad (\text{A.7})$$

For every fixed nonzero direction $v \in [0, \infty)^n$, set $\mathcal{F}_v(t) = \Pi(tv)^2 \mathcal{A}(tv)$. Then

$$\frac{d \log \mathcal{F}_v(t)}{d \log t} < -\frac{3}{100} \quad \left(t \sum_i v_i \geq \frac{16}{9}\right).$$

In particular, $\mathcal{F}_v$ is strictly decreasing on this range. Equality in (A.6) holds exactly when two odds are positive and equal to $s/2$, up to zero coordinates and permutation.

Proof. We first prove the intermediate-range envelope. After relabeling, isolate $x = o_1$ and $y = o_2$, and write

$$a = x + y, \quad k = xy, \quad 0 < k \leq \frac{a^2}{4}.$$

For the untouched coordinates, set

$$H = \sum_{j=3}^n g(o_j), \quad \Pi_0 = \prod_{j=3}^n (1 + o_j)^{-1}.$$

With

$$z_a(k) = \frac{1}{(1+a+k)^2}, \quad R_a(k) = z_a(k)\{g(x) + g(y)\},$$

the full functional retains the untouched coordinates exactly:

$$\Pi^2 \mathcal{A} = \Pi_0^2 \{R_a(k) + H z_a(k)\}. \quad (\text{A.8})$$

Put $D = 1 + a^2 - 2k + k^2$. Direct symmetric expansion gives

$$R_a(k) = \frac{N(a, k)}{D^2(1 + a + k)^2},$$

where

$$\begin{aligned} N(a, k) &= a + 2a^2 + a^3 + (a^3 - a - 4)k \\ &\quad + (2a^2 - a + 8)k^2 + (a - 4)k^3. \end{aligned}$$

The pair endpoints are a merge and an equal split. Their values are

$$\mathcal{C}_1(a) = \frac{a}{(1 + a^2)^2}, \quad \mathcal{C}_2(a) = \frac{a}{(1 + a/2)^2(1 + a^2/4)^2},$$

and

$$\mathcal{C}_2(a) - \mathcal{C}_1(a) = -\frac{a^2(a^2 - 6a + 4)(a^3 + 10a^2 + 4a + 16)}{(a + 2)^2(a^2 + 1)^2(a^2 + 4)^2}.$$

Thus the endpoints exchange order at $a_* = 3 - \sqrt{5}$.

Two exact Bernstein certificates control the whole pair curve. For $0 \leq a \leq a_*$, define

$$\mathcal{S}_{\text{raw}}(a, k) := \frac{aD^2(1 + a + k)^2 - (1 + a^2)^2 N(a, k)}{k}, \quad \mathcal{S}(a, k) := \frac{\mathcal{S}_{\text{raw}}(a, k)}{(1 + a)^2}.$$

Both quotients are polynomials, and

$$R_a(0) - R_a(k) = \frac{k\mathcal{S}_{\text{raw}}(a, k)}{(1 + a^2)^2 D^2(1 + a + k)^2}.$$

Under

$$a = a_* r, \quad k = \frac{a_*^2 t}{4}, \quad (r, t) \in [0, 1]^2,$$

the uncanceled mapped quotient has monomial bidegree $(13, 5)$. Cancelling the positive common factor $(1 + a)^2$ gives a reduced quotient of bidegree $(11, 5)$. Their degree- $(60, 20)$ Bernstein tables differ but have the same sign pattern: exactly 1280 entries are positive, and the sole zero is the corner coefficient $(60, 20)$. Hence

$$R_a(k) \leq R_a(0),$$

strictly for $k > 0$ except at $(a, k) = (a_*, a_*^2/4)$. Since z_a decreases, the same merge inequality holds after the term $H z_a(k)$ in (A.8) is added.

For $a_* \leq a \leq 16/9$, define $\theta \in [0, 1]$ by

$$z_a(k) = \theta z_a(0) + (1 - \theta) z_a(a^2/4).$$

Define $\mathcal{L}(a, k)$ by the exact identity

$$\begin{aligned} &\theta R_a(0) + (1 - \theta) R_a(a^2/4) - R_a(k) \\ &= \frac{k(a^2 - 4k)\mathcal{L}(a, k)}{(1 + a^2)^2(4 + a^2)^2(1 + a + k)^2(8 + 8a + a^2)D^2}. \end{aligned}$$

Under

$$a = a_* + \left(\frac{16}{9} - a_*\right)r, \quad k = \frac{a^2 t}{4},$$

the remaining quotient has bidegree $(13, 4)$ and all 70 tensor Bernstein coefficients are positive. Therefore

$$R_a(k) < \theta R_a(0) + (1 - \theta)R_a(a^2/4) \quad \left(0 < k < \frac{a^2}{4}\right). \quad (\text{A.9})$$

The untouched term $H z_a(k)$ is affine in the same coordinate, so (A.9) holds for the complete pair profile, uniformly in $H \geq 0$.

Fix $s \in [1, 16/9]$ and maximize $\Pi^2 \mathcal{A}$ on the closed odds simplex. No positive pair in a maximizer can have sum at most a_* . For $a < a_*$, merging the pair strictly increases the objective. At $a = a_*$, this is also true for an equal pair when $H > 0$; if $H = 0$, then $s = a_* < 1$. Every remaining pair lies in the strict chord region, so it must be equal. Thus all positive odds in any maximizer are equal. If their number is $d \geq 2$, then

$$d < \frac{2s}{a_*} \leq \frac{8}{9}(3 + \sqrt{5}) < 5.$$

The case $d = 1$ is already in the resulting list, so in all cases $d \in \{1, 2, 3, 4\}$. The equal-block value is

$$\mathcal{C}_d(s) = \frac{s}{(1 + (s/d)^2)^2 (1 + s/d)^{2(d-1)}}.$$

After positive factors are removed, the three differences $\mathcal{C}_2 - \mathcal{C}_d$, $d = 1, 3, 4$, have the respective factors

$$-(s^2 - 6s + 4), \quad 8s^3 - 33s^2 - 18s + 108, \quad s^4 + 12s^3 - 64s^2 + 256.$$

They are strictly positive on the intermediate range. The first polynomial inside the minus sign decreases there and starts at -1 . The second decreases there and has right-endpoint value $12140/729$. The third is at least $256 - 64(16/9)^2 > 0$. This proves (A.6) and its equality statement.

For the small range, expand term by term:

$$\Pi^2 \mathcal{A} = \sum_i \frac{o_i}{(1 + o_i^2)^2 \prod_{j \neq i} (1 + o_j)^2}.$$

If $r_i = s - o_i$, then the product over $j \neq i$ is at least $1 + r_i$. The elementary inequality

$$(1 + x^2)^2 (1 + y)^2 \geq 3(x + y) \quad (x > 0, y \geq 0) \quad (\text{A.10})$$

therefore bounds every summand with $o_i > 0$ by $o_i/(3s)$; a zero-odds summand vanishes. To verify (A.10), put $A = (1 + x^2)^2$. The difference is

$$Ay^2 + (2A - 3)y + (A - 3x).$$

Its constant term is positive because $\min_{x>0} A/x = 16/(3\sqrt{3}) > 3$. If $2A - 3 < 0$, then $x < 1/2$, and four times the leading coefficient times the quadratic minimum is $12A(1 - x) - 9 > 0$, since $A(1 - x) \geq 25/32$. This proves (A.5).

It remains to prove the large-range envelope. Discard the zero coordinates of a fixed nonnegative direction. Along the resulting positive ray $o_i = tv_i$, let $\mathcal{F}(t) = \Pi(tv)^2 \mathcal{A}(tv)$ and put

$$e(o) = \frac{og'(o)}{g(o)}.$$

Logarithmic differentiation gives

$$\frac{d}{d \log t} \log \mathcal{F}(t) = \frac{\sum_i g(o_i) e(o_i)}{\mathcal{A}} - 2 \sum_i \frac{o_i}{1 + o_i}.$$

The exact identities

$$\begin{aligned} \frac{5}{4} - e(o) &= \frac{9o^3 + 17o^2 - 7o + 1}{4(1+o)(1+o^2)} > 0, \\ 17o^2 - 7o + 1 &= 17 \left(o - \frac{7}{34} \right)^2 + \frac{19}{68}, \end{aligned}$$

and

$$\frac{x}{1+x} + \frac{y}{1+y} - \frac{x+y}{1+x+y} = \frac{xy(x+y+2)}{(1+x)(1+y)(1+x+y)} > 0$$

imply

$$\frac{d}{d \log t} \log \mathcal{F}(t) < \frac{5}{4} - \frac{2s}{1+s} \leq -\frac{3}{100} \quad \left(s \geq \frac{16}{9} \right).$$

Scale back to total odds 16/9 and apply (A.6). This gives (A.7).

The two tables are generated by `verify/finite/support_odds/exact_certificate.py`; the independent basis conversion is in the corresponding `independent_certificate.py`. Equation (7.1) specifies the conversion, and the programs check every coefficient sign. Their source hashes and the reported sign counts are in `certificates/finite/results.json`. $\square$

Lemma A.3 (Endpoint atom comparison). *Under the hypotheses and notation of Lemma A.1, the closed deficit at the lower support and the strict deficit at the upper support are strictly positive.*

Proof. Normalize $V = 1$, put $o_i = p_i/q_i$, and set

$$w_i = p_i q_i h_i^2, \quad \sum_i w_i = 1.$$

The distance to the lower support, its atom, and the cubic sum are

$$x_0 = \sum_i \sqrt{o_i w_i}, \quad \Pi = \prod_i (1 + o_i)^{-1},$$

$$\mathcal{B}_h = \sum_i \frac{1 + o_i^2}{\sqrt{o_i}(1 + o_i)} w_i^{3/2}.$$

Use s and $\mathcal{A}$ from (A.4). With

$$a_i = \frac{o_i}{(1 + o_i)^2}, \quad r_i = \frac{1 + o_i^2}{(1 + o_i)^2}, \quad z_i = \sqrt{a_i} h_i,$$

we have $\sum_i z_i^2 = 1$ and $\mathcal{B}_h = \sum_i (r_i / \sqrt{a_i}) z_i^3$. Cauchy–Schwarz, twice, gives

$$1 \leq \mathcal{B}_h \sum_i \frac{\sqrt{a_i}}{r_i} z_i \leq \mathcal{B}_h \sqrt{\mathcal{A}}, \quad x_0^2 \leq s.$$

Let $Q(x) = 1 - \Phi(x)$. Since Q decreases, it is enough to prove

$$\frac{C_E}{\sqrt{\mathcal{A}}} + Q(\sqrt{s}) > \Pi. \tag{A.11}$$

Put

$$f(s) := (1+s)Q(\sqrt{s}), \quad f'(s) < 0 \quad (s > 0). \quad (\text{A.12})$$

To prove the derivative sign, take $x = \sqrt{s}$, $M(x) = Q(x)/\phi(x)$, and $u(x) = (1+x^2)/(2x)$, the required derivative sign is $M(x) < u(x)$. For $x \geq 1$ this follows from the Mills bound $M(x) < 1/x$. On $(0, 1)$, put $d = u - M$. We have $d(0+) = \infty$ and $d(1) > 0$. Since $M' = xM - 1$, at any interior critical point of d ,

$$M(x) = \frac{3x^2 - 1}{2x^3}, \quad d(x) = \frac{(1-x^2)^2}{2x^3} > 0.$$

Thus d cannot have a nonpositive minimum, proving (A.12).

First suppose $0 < s \leq 1$. By (A.5),

$$\frac{C_E}{\Pi\sqrt{\mathcal{A}}} > \frac{7}{10},$$

using $C_E > 49/120$ and $\sqrt{3} > 12/7$. Also $\Pi \leq (1+s)^{-1}$. The alternating exponential bound on $[0, 1]$ and $\phi(0) < 2/5$ give

$$Q(1) > \frac{1}{2} - \frac{2}{5} \left(1 - \frac{1}{6} + \frac{1}{40}\right) = \frac{47}{300}.$$

By (A.12),

$$\frac{Q(\sqrt{s})}{\Pi} \geq (1+s)Q(\sqrt{s}) \geq 2Q(1) > \frac{47}{150} > \frac{3}{10}.$$

The two strict bounds prove (A.11) in the small range.

For $1 \leq s \leq 16/9$, write $u = s/2$. The intermediate-range envelope gives

$$\frac{C_E}{\Pi\sqrt{\mathcal{A}}} \geq C_E \frac{(1+u)(1+u^2)}{\sqrt{2u}} > \frac{49}{64}. \quad (\text{A.13})$$

The factor after C_E increases for $u \geq 1/2$; its logarithmic derivative has the sign of $5u^3 + 3u^2 + u - 1 > 0$. At $s = 16/9$, the alternating bound

$$e^{-t^2/2} \leq 1 - \frac{t^2}{2} + \frac{t^4}{8} - \frac{t^6}{48} + \frac{t^8}{384} \quad \left(0 \leq t \leq \frac{4}{3}\right)$$

and $\phi(0) < 2/5$ give

$$Q\left(\frac{4}{3}\right) > \frac{16727087}{186004350}, \quad \frac{25}{9}Q\left(\frac{4}{3}\right) - \frac{15}{64} > \frac{33055039}{2142770112} > 0.$$

Consequently,

$$\frac{Q(\sqrt{s})}{\Pi} \geq (1+s)Q(\sqrt{s}) > \frac{15}{64}. \quad (\text{A.14})$$

The bounds in (A.13) and (A.14) again sum to more than one.

Finally, if $s \geq 16/9$, Proposition A.2 and the exact margin

$$\frac{2401}{14400} - \frac{944784}{6076225} = \frac{7873013}{699981120} > 0$$

give $\Pi^2\mathcal{A} < C_E^2$. The cubic term alone proves (A.11). Hence, in every range,

$$C_E\mathcal{B}_h + Q(x_0) - \Pi \geq \frac{C_E}{\sqrt{\mathcal{A}}} + Q(\sqrt{s}) - \Pi > 0.$$

This is the closed deficit at the lower support. Complementing every B_i preserves V and $\mathcal{B}_h$ and carries it to the strict deficit at the upper support. $\square$

Proof of Proposition 3.2. The two preceding endpoint lemmas supply the two signs at each endpoint. $\square$

B The scalar Fourier inequality

We prove the scalar input (4.3) used in the atom and interval estimates. Write

$$I_0(W) = \frac{1}{\pi} \int_0^\pi e^{W \cos \theta} d\theta, \quad I_1(W) = I_0'(W).$$

For $W = \nu c/4$, the inequality $1 - u \leq e^{-u}$ gives

$$\mathcal{M}(\nu, c) \leq \sqrt{2\pi W} e^{-W} I_0(W). \quad (\text{B.1})$$

Proposition B.1 (Analytic ranges of the scalar atom bound). *The scalar inequality (4.3) holds in each of the following ranges:*

$$1 \leq \nu \leq 2, \quad 0 \leq t < 1; \quad \nu \geq 1, \quad 0 \leq t \leq \frac{1}{\sqrt{10}}; \quad \nu \geq 1, \quad t_0 \leq t < 1,$$

where

$$t_0 := t_* + \sqrt{\frac{1}{5K_E}} = 0.6035634 \dots$$

The only region not covered by these analytic estimates is

$$\nu > 2, \quad \frac{1}{\sqrt{10}} < t < t_0. \quad (\text{B.2})$$

For each fixed $t < 1$, $\mathcal{M}(\nu, 1-t^2) \rightarrow 1$ as $\nu \rightarrow \infty$. Across the full scalar domain, the limiting margin vanishes only at $t = t_*$. This point lies in the already proved range $t < 1/\sqrt{10}$, not in the closure of the remaining strip (B.2); the latter has positive limiting margin at least $K_E(1/\sqrt{10} - t_*)^2$.

Proof. Put $c = 1 - t^2$ and $U(\theta) = (1 - c \sin^2(\theta/2))^{1/2}$, with normalized Lebesgue measure on $[0, \pi]$. If $1 \leq \nu \leq 2$, Lyapunov's inequality gives

$$\mathbb{E}U^\nu \leq (\mathbb{E}U^2)^{\nu/2} = (1 - c/2)^{\nu/2}.$$

Maximizing the resulting squared bound over $c \in [0, 1]$ gives

$$\mathcal{M}(\nu, c)^2 \leq \frac{\pi\nu}{\nu+1} \left(\frac{\nu}{\nu+1} \right)^\nu \leq \frac{8\pi}{27} < 1.$$

The middle expression is increasing on $[1, 2]$, and $\pi < 27/8$ proves the last inequality. This proves the first range.

Suppose next that $9/10 \leq c < 1$. For $x \in [0, \pi/2]$, set

$$D_c(x) = -\log(1 - c \sin^2 x) - cx^2.$$

The derivative of $H_c(x) = \sin(2x)/(1 - c \sin^2 x) - 2x$ is

$$H'_c(x) = -\frac{2 \sin^2 x \{c^2 \sin^2 x - 3c + 2\}}{(1 - c \sin^2 x)^2}.$$

The factor in braces changes sign once. Since $H_c(0) = 0$ and $H_c(\pi/2) = -\pi$, the function D_c first increases and then decreases, so its minimum occurs at an endpoint. The endpoint $x = 0$ gives zero, while

$$D_c(\pi/2) = -\log(1 - c) - \frac{c\pi^2}{4} \geq 0.$$

It suffices to check $c = 9/10$, using $\log 10 > 23/10$ and $\pi < 22/7$; the endpoint expression then increases with c . Therefore $1 - c \sin^2(\theta/2) < e^{-c\theta^2/4}$ when $0 < \theta \leq \pi$. At $c = 1$, the same strict comparison follows from $\tan x > x$ for $0 < x < \pi/2$. Integrating over the original finite interval gives

$$\mathcal{M}(\nu, c) < \operatorname{erf}\left(\pi\sqrt{\frac{\nu c}{8}}\right) < 1 \quad (\nu > 0).$$

This proves the second range.

For the third range put $W = \nu c/4$. Equation (B.1) gives

$$\mathcal{M}(\nu, c) \leq e^{-W} I_0(W) \sqrt{2\pi W} < \frac{6}{5}.$$

For $0 < W \leq 1$, the series inequality $I_0(W) \leq e^{W^2/4}$ reduces the claim to the increasing function $\sqrt{2\pi W} e^{-W+W^2/4}$ and its value at one. For $W \geq 1$, let $r = I_1/I_0$. Termwise comparison gives $r(1) < 1/2$. The Riccati identity $r' = 1 - r^2 - r/W$ then prevents r from crossing $1 - 1/(2W)$, because at a contact its derivative is $1/(4W^2)$ whereas the comparison function has derivative $1/(2W^2)$. Thus the displayed Bessel expression decreases for $W \geq 1$. Finally, $\sqrt{2\pi} e^{-3/4} < 6/5$ follows from $\pi < 22/7$ and the first five positive terms of $e^{3/2}$. When $t \geq t_0$, the right side of the scalar inequality (4.3) is at least $6/5$.

For fixed $c > 0$, the final limit follows from the change of variables $\theta = \sqrt{8/(\nu c)} u$ and dominated convergence in the resulting Laplace integral. The limiting right side is $1 + K_E(t - t_*)^2$, whose gap from one vanishes only at $t = t_*$. $\square$

Proposition B.2 (Full scalar atom bound). *The scalar inequality (4.3) holds for every $\nu \geq 1$ and $0 \leq t < 1$, strictly at every finite point. Its margin can tend to zero only when $\nu \rightarrow \infty$ and $t \rightarrow t_*$.*

Proof. Proposition B.1 leaves only the strip in the scalar region (B.2). Put

$$a = \frac{1}{\sqrt{10}}, \quad c = 1 - t^2, \quad J_\nu(t) = \frac{2}{\pi} \int_0^{\pi/2} (\cos^2 \theta + t^2 \sin^2 \theta)^{\nu/2} d\theta.$$

For integers $m \geq 0$, beta integration gives

$$q_m(t) := J_{2m}(t) = 4^{-m} \sum_{k=0}^m \binom{2k}{k} \binom{2m-2k}{m-k} t^{2k}. \quad (\text{B.3})$$

If $2m - 2 \leq \nu \leq 2m$, $m \geq 2$, Hölder log-convexity and $\lambda = (\nu - 2m + 2)/2$ give

$$J_\nu(t) \leq q_{m-1}(t)^{1-\lambda} q_m(t)^\lambda.$$

Set $\ell_m(t) = \log(q_{m-1}(t)/q_m(t)) > 0$. Maximizing the resulting factor $\sqrt{\nu} \exp\{-\nu \ell_m(t)/2\}$ over all positive real ν , a larger domain than the order slab, gives

$$\mathcal{M}(\nu, 1 - t^2) \leq U_m(t) := \sqrt{\frac{\pi(1-t^2)}{2e\ell_m(t)}} q_{m-1}(t) e^{(m-1)\ell_m(t)}. \quad (\text{B.4})$$

It remains to bound finitely many one-variable functions. Divide $[0, 1]$ into the 2048 exact rational intervals $[j/2048, (j+1)/2048]$ and map them affinely onto $[a, t_0]$. Outward Arb evaluation of the positive polynomial (B.3) and the slab majorant (B.4) proves

$$1 + K_E(t - t_*)^2 - U_m(t) > \frac{19}{1000}. \quad (\text{B.5})$$

The compact gap (B.5) holds on every such interval and for every $2 \leq m \leq 64$. These $2048 \cdot 63 = 129024$ boxes cover $2 \leq \nu \leq 128$ and $a \leq t \leq t_0$, including all four edges. The recorded smallest outward lower endpoint exceeds 0.0194747706542. The compact interval proof and independent Arb implementation are, respectively,

```
verify/finite/scalar_atom/compact_certificate.py
verify/finite/scalar_atom/independent_certificate.py
```

Their digests and replay commands are recorded in `certificates/finite/`.

For $\nu \geq 128$, the Bessel majorant from the proof of Proposition B.1 is decreasing because $W = \nu c/4 \geq 1$. Throughout the remaining strip,

$$W \geq 32(1 - t_0^2).$$

Since $t \geq a > t_*$, the right side of the scalar inequality is at least $1 + K_E(a - t_*)^2$. Since the Bessel majorant decreases in W , its left side is at most its value at $W = 32(1 - t_0^2)$. Thus one outward evaluation gives

$$1 + K_E(a - t_*)^2 - \sqrt{2\pi W} e^{-W} I_0(W) > 0.0180176 \quad (W = 32(1 - t_0^2)).$$

This proves the tail for every real order, and hence the full scalar inequality. All finite-region bounds are strict. All regions outside the small-skew range have the fixed positive margins just proved. In the small-skew range, the preceding error-function estimate gives

$$1 + K_E(t - t_*)^2 - \mathcal{M}(\nu, 1 - t^2) > K_E(t - t_*)^2 + \operatorname{erfc} \left(\pi \sqrt{\frac{\nu(1 - t^2)}{8}} \right).$$

Hence a vanishing-margin sequence must have $t \rightarrow t_*$ and $\nu \rightarrow \infty$; the Laplace limit in Proposition B.1 shows that this boundary is attained as a limit. $\square$

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